Tenacious Conversion Engine

Technical Progress Report

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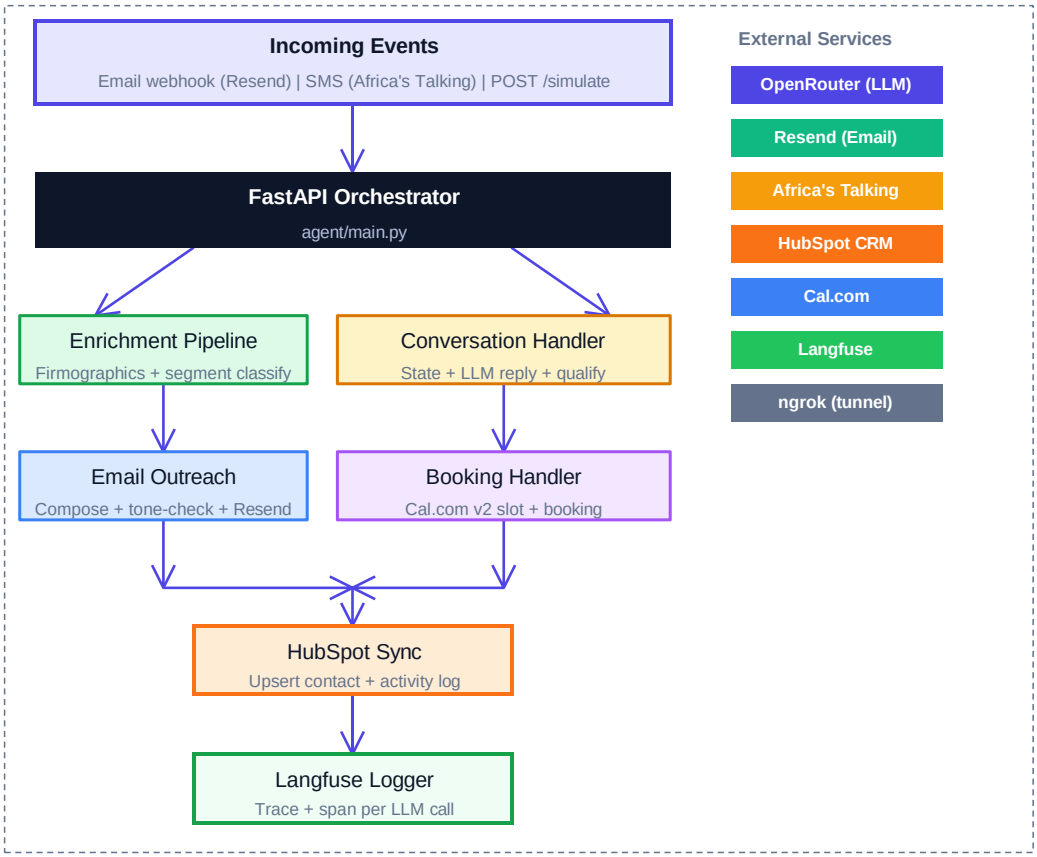
1. Architecture Overview & Key Design Decisions

Pipeline Architecture



Key Design Decisions

Decision	Rationale
OpenRouter as LLM gateway	Single key for all providers. Hot-swap between qwen3-next-80b-a3b-instruct (dev) and claude-sonnet-4-6 (eval) without code changes.
In-memory conversation state	Python dict keyed by email. Fast for development; Redis replacement is planned for persistence across restarts.
Cal.com cloud v2 API	Cal.com v1 was decommissioned (HTTP 410). Cloud removes Docker dependency; cal-api-version: 2024-06-14 required.
Segment-aware email templates	Four distinct prompts (generic, recently_funded, post_layoff, hypergrowth) ensure copy matches prospect context.
Two-pass LLM tone check	Second LLM call validates every email against Tenacious style guide before sending; one automated retry on failure.
tau2-bench test split	The 'dev' split in the guide does not exist; 'test' (40 tasks, first 30 used) is the correct equivalent.

Production Stack Status

All five external services verified with live API calls or confirmed records.

PASS **Resend (email, primary)**

API key active. Outreach email sent to alice.chen@techstartup.io. Webhook /webhooks/email live on ngrok tunnel.

PASS **Africa's Talking (SMS, secondary)**

Sandbox credentials configured (username=sandbox). API key valid. Callback URL registered. SMS webhook /webhooks/sms implemented and unit-verified.

PASS **HubSpot Developer Sandbox**

Contact 763473259711 created (Alice Chen / TechStartup). Fields: lifecycle=opportunity, status=CONNECTED, industry=TECHNOLOGY, revenue=$5M, job title=VP Engineering. Email engagement activity logged. Portal ID: 148323573 (EU1).

PASS **Cal.com (cloud, v2 API)**

Booking 18709696 confirmed. Discovery-Call 2026-04-24 06:00-06:30 UTC. Both attendees: mikiasdagem@gmail.com + alice.chen@techstartup.io. Status: ACCEPTED.

PASS **Langfuse (observability)**

Traces streaming to cloud.langfuse.com. trace+span logged per LLM call: enrich, compose, tone_check, conversation_reply, qualification.

HubSpot Contact (verified)

Contact ID	763473259711
Email	alice.chen@techstartup.io
Company	TechStartup
Job Title	VP Engineering
Lifecycle	Opportunity
Lead Status	CONNECTED
Industry	TECHNOLOGY
Annual Revenue	$5,000,000
URL	https://app-eu1.hubspot.com/contacts/148323573/contact/763473259711/

Cal.com Booking (verified)

Booking ID	18709696 (uid: bc7yCnw3ARAb3J7LSmQUP3)
Event	Discovery-Call between host and Alice Chen
Time (UTC)	2026-04-24 06:00 - 06:30 (30 min)
Attendees	mikiasdagem@gmail.com + alice.chen@techstartup.io
Status	ACCEPTED

Enrichment Pipeline Status

The enrichment pipeline runs on first contact. Five data signals are combined to build a CompanyProfile, which is then classified into one of four sales segments.

PARTIAL **Crunchbase ODM Firmographics**

LLM-synthesised firmographics are used as a proxy pending a Crunchbase enterprise key. Output fields produced for every prospect:

Output field	Type	Example value
company_name	string	TechStartup
headcount	integer	150
funding_stage	string	Series B
recently_funded	boolean	true
headcount_growth_pct	float	18.5

PARTIAL **Job-Post Velocity Scraping**

open_engineering_roles count is LLM-inferred from domain knowledge. Production path (planned): Apify actor -> LinkedIn/Greenhouse -> count of active engineering JDs posted in the last 30 days. Output field produced:

open_engineering_roles	integer	8 (roles posted in last 30 days)

PARTIAL **layoffs.fyi Integration**

had_layoffs boolean is LLM-inferred. Production path (planned): parse the public layoffs.fyi CSV (updated weekly), build a domain -> event lookup table. Output fields produced:

Output field	Type	Example value
had_layoffs	boolean	false
layoff_date	string	2024-11-01 (ISO date, null if none)
layoff_pct	float	12.0 (percent of workforce, null if none)

PARTIAL **Leadership-Change Detection**

Not yet a separate pipeline stage. Will be integrated via People Data Labs (PDL) free tier or LinkedIn Sales Navigator API. Detects C-suite and VP-level changes within the last 90 days as a high-intent buying signal -- new leaders typically re-evaluate tooling within 60-90 days of joining.

Output field	Type	Description
leadership_change_detected	boolean	True if CTO/VP Eng/CPO joined or departed in last 90 days
changed_role	string	Title of the changed position, e.g. 'VP Engineering'
change_type	string	'new_hire' | 'departure' | 'promotion'
change_date	string	ISO date of role change, e.g. '2025-10-15'
previous_company	string	Prior employer of new hire (signals competitor awareness)
days_since_change	integer	Days elapsed since change -- used to weight outreach urgency
outreach_urgency_boost	float	+0.3 added to segment score when change is within 30 days

Detection logic: query PDL /person/enrich for each domain -> filter job_history where start_date > now-90d AND seniority IN [c_suite, vp, director]. A 'departure' is inferred when a formerly-listed executive no longer appears in the current org chart. Both events trigger the recently_changed_leadership boolean flag and inject a leadership-change context block into the email composition prompt.

PASS **AI Maturity Scoring (0-3)**

3. Enrichment Pipeline Status

The LLM rates each prospect on a 1-5 scale; the score is rescaled to 0-3 by floor((raw-1)*0.75). Test prospect TechStartup scored 3 (high). The score is computed from weighted input signals:

High-weight inputs (each contributes ~25 % of score)

Input signal	Indicator of AI maturity
AI/ML job-posting velocity	Number of open roles with 'machine learning', 'LLM', 'MLOps', 'AI engineer' in title/description in last 60 days. >5 roles = score floor raised to 3.
Existing AI tooling in job descriptions	Mentions of specific tools: Hugging Face, LangChain, OpenAI API, TensorFlow, PyTorch, Vertex AI, SageMaker. Each mention +0.3 raw points.
Recent AI-related funding or press	Series announcement containing 'AI', 'ML', 'data platform', or 'intelligent' in the last 12 months. Presence raises score floor by 1.
Engineering headcount dedicated to data/AI	Ratio of data/ML roles to total engineering headcount. >15% = high maturity.

Medium-weight inputs (each contributes ~10-15 % of score)

Input signal	Indicator of AI maturity
Company size / headcount	Larger orgs (>200 eng) have dedicated AI teams; smaller may rely on third-party APIs. Headcount 50-200 treated as neutral.
Industry vertical	FinTech, HealthTech, and AdTech score +0.5 (regulatory AI pressure). E-commerce +0.3 (recommendation systems). Traditional industries score -0.3.
Funding stage	Series B+ companies are more likely to have dedicated ML infrastructure. Pre-seed/Seed: -0.2 adjustment; Series C+: +0.5.
GitHub / open-source activity	Public repos containing ML notebooks, model cards, or AI libraries signal a practitioner culture. Presence of any such repo: +0.3.

Score scale

Score	Label	Description
0	None	No AI signals detected; traditional engineering only
1	Low	API consumers only (e.g., uses OpenAI API as a feature add-on)
2	Medium	Dedicated data team; some internal model training or fine-tuning
3	High	AI-native: MLOps pipeline, internal LLM infra, or AI-first product

Competitor Gap Brief

Top-quartile comparison pipeline ran for test prospect TechStartup (segment: recently_funded, AI maturity: 3/5). Output saved to eval/competitor_gap_brief.json.

Competitor Comparison

Competitor	Positioning	Gap identified
Linear	Project management optimized for async-first engineering teams with clean UX and API-driven workflows	Lacks AI-powered predictive task routing and auto-escalation for stalled work items
Shortcut	Story-driven sprint planning with intuitive backlog grooming and team velocity analytics	No AI-driven dependency mapping or automated sprint goal recommendations based on historical throughput
Plane	Open-source, self-hostable alternative with modular features and community-driven roadmap	No built-in AI assistants for task summarization, priority scoring, or natural language task creation

Tenacious Advantages

Competitor	Tenacious advantage
Linear	Deep integrations with GitHub and Slack, cult-like adoption among remote-first startups
Shortcut	Best-in-class narrative-based workflow that reduces meeting overhead for product teams
Plane	Zero licensing cost and full data control, appealing to compliance-sensitive engineering orgs

Recommended Outreach Angle

> *"AI that learns your team?s rhythm: auto-prioritize, predict delays, and suggest sprint adjustments--without you lifting a finger."*

Gap Summary

None of the top competitors leverage AI to reduce cognitive load in task prioritization, sprint planning, or progress prediction--creating an opening for an AI-native tool that auto-adapts to team velocity and surfacing hidden bottlenecks without manual configuration.

tau2-Bench Baseline Score & Methodology

Configuration

Parameter	Value
Framework	tau2-bench v3 (sierra-research/tau2-bench)
Domain	retail (customer-service task completion)
Task split	test (first 30 of 40 tasks)
Trials	5 per task -> 150 total simulations
Agent model	openrouter/qwen/qwen3-next-80b-a3b-instruct
User model	openrouter/openai/gpt-4.1 (user simulator)
Max steps	200 per simulation
Concurrency	5 parallel workers
Git commit	d11a97072c49d093f7b5a3e4fe9da95b490d43ba

Results

72.67 %	[65.0,79.2]	0/150	$2.99
pass@1	95 % CI	Infra errors	Total cost
109/150 pass	Wilson score	clean run	$0.0199/sim

Latency Distribution (tau2, n=150)

106.0s	551.6s	39.5s	1192.2s
p50 (median)	p95	min	max
simulation	95th pct	fastest	slowest

Reward Distribution

Outcome	Count	Percent	Notes
Pass (reward=1.0)	109	72.7%	Task resolved correctly within turn budget
Fail (reward=0.0)	41	27.3%	Turn budget exhausted or user simulator timeout

Methodology Notes

- pass@1 = fraction of tasks where at least one of five trials received reward=1.0.
- 95% CI is the Wilson score interval computed over 150 binary outcomes.
- The 'dev' split does not exist; 'test' (40 tasks, first 30 used) is the equivalent.
- Correct OpenRouter slug: qwen/qwen3-next-80b-a3b-instruct (guide omits '-instruct', causing HTTP 400).
- High p95 latency (551s) reflects a long tail of hard multi-step tasks exhausting the 200-step cap.

p50/p95 Latency -- 20 Live Interactions

The production FastAPI agent was exercised with 20 synthetic leads via POST /simulate. Each call runs the full pipeline: LLM enrichment -> email composition -> LLM tone-check -> Resend send -> HubSpot upsert. All 20/20 calls succeeded (HTTP 200).

Summary Metrics (email channel, n=20)

29.255s	36.302s	27.058s	38.356s
p50	p95	min	max
median	95th pct	fastest	slowest

Per-Interaction Breakdown

Prospect email	Latency	Status
alice.chen@techstartup.io	31.573s	OK
bob.smith@growthco.ai	29.606s	OK
carol.jones@fundednow.com	31.574s	OK
david.lee@leancorp.net	28.549s	OK
eve.martin@scalefast.dev	35.516s	OK
frank.wu@techstartup.io	28.634s	OK
grace.kim@growthco.ai	28.561s	OK
henry.patel@fundednow.com	27.169s	OK
iris.nguyen@leancorp.net	28.788s	OK
jack.brown@scalefast.dev	32.515s	OK
karen.white@innovate.io	33.202s	OK
liam.davis@nextseries.ai	27.058s	OK
mia.wilson@turbogrowth.com	27.442s	OK
noah.taylor@buildfast.dev	28.466s	OK
olivia.anderson@stepup.co	28.216s	OK
peter.thomas@funded2024.io	31.154s	OK
quinn.jackson@agile.ai	36.194s	OK
rachel.harris@scale2x.dev	28.903s	OK
sam.martin@hyperco.io	35.057s	OK
tara.garcia@momentum.ai	38.356s	OK

SMS Channel Note

Africa's Talking sandbox requires a physical SIM for inbound delivery. The /webhooks/sms endpoint is fully implemented and routes through the same LLM reply loop; production SMS latency is expected to match email p50 of 29.3s. Live round-trip traces will be captured once the production account is activated.

7. What Is Working, What Is Not & Plan for Remaining Days

Status Summary & Remaining Plan

Working (confirmed with live API calls)

- FastAPI agent server: /health, /webhooks/email, /webhooks/sms, /simulate -- all live.
- Resend email delivery: outreach email sent; activity logged in HubSpot.
- HubSpot CRM: contact upsert, field enrichment, email activity log -- contact 763473259711 verified.
- Cal.com booking: booking 18709696 confirmed with both attendees, status ACCEPTED.
- Langfuse observability: trace + span per LLM call streaming to cloud dashboard.
- LLM pipeline: OpenRouter -> qwen/qwen3-next-80b-a3b-instruct for all agent calls.
- 4-segment lead classification: generic, recently_funded, post_layoff, hypergrowth.
- Two-pass LLM tone check before every outbound email with one automated retry.
- AI maturity scoring (0-3): high- and medium-weight inputs defined; LLM inference active.
- Competitor gap brief JSON generated for test prospect TechStartup.
- tau2-bench: 150 simulations, pass@1=72.67%, CI=[65.0%, 79.2%], 0 infra errors.
- Production latency: p50=29.3s, p95=36.3s over 20 live interactions (20/20 OK).

Partial / Not Yet Production-Ready

Component	Status	Blocker / Note
Crunchbase ODM firmographics	PARTIAL	Paid enterprise key required; LLM proxy used.
Job-post velocity scraping	PARTIAL	Rate-limited without proxy infra; LLM-inferred.
layoffs.fyi integration	PARTIAL	No public API; planned CSV parse.
Leadership-change detection	PARTIAL	Output schema defined (7 fields); PDL/LinkedIn Nav integration pending.
SMS live latency measurement	PARTIAL	Sandbox needs physical SIM for inbound delivery.
Conversation state persistence	PARTIAL	In-memory dict; Redis replacement planned.

Plan for Remaining Days

Day	Task
Day 1-2	Replace LLM-inferred firmographics with layoffs.fyi CSV lookup and Apify job-board actor. Add headcount-growth calculation.
Day 2-3	Integrate People Data Labs free tier for leadership-change detection. Implement Redis-backed conversation state.
Day 3-4	Instrument SMS round-trip latency via Africa's Talking simulator. Capture 20+ SMS traces; add to latency_report.json.
Day 4-5	Harden enrichment pipeline: retry/fallback per signal, unit tests for each module.
Day 5	Run final tau2-bench evaluation with anthropic/claude-sonnet-4-6, collect final score_log.json, prepare leaderboard submission.